

The symbol you're describing, $\binom{n}{2}$, is known as a **binomial coefficient**. It represents the number of ways to choose 2 items from a set of n items without regard to the order of selection. Mathematically, it's calculated as:

$$\binom{n}{2} = \frac{n!}{2!(n-2)!}$$

where $n!$ (n factorial) is the product of all positive integers up to n . For example, $\binom{5}{2}$ would be:

$$\binom{5}{2} = \frac{5!}{2!(5-2)!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{2 \times 1 \times 3 \times 2 \times 1} = 10$$

This means there are 10 ways to choose 2 items from a set of 5.